

Day_4_Visulization with matplotlib

September 10, 2026

1 Day 4 — Visualization with Matplotlib

1.1 Objective

The objective of this task is to visualize the Customer Churn dataset using Matplotlib.

Different types of plots such as Line Plot, Bar Plot, Histogram, and Scatter Plot are used to understand trends, comparisons, distributions, and relationships between numerical variables.

```
[7]: import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel(
    "Day_4_Ensemble_Learning_Customer_Churn.xlsx",
    sheet_name="Customer Churn Data"
)

df
```

```
[7]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges	Contract	\
0	1001	25	3	65	Month-to-month	
1	1002	34	18	72	One year	
2	1003	45	36	85	Two year	
3	1004	29	7	55	Month-to-month	
4	1005	52	48	95	Two year	
5	1006	41	24	78	One year	
6	1007	23	2	50	Month-to-month	
7	1008	38	30	88	One year	
8	1009	49	42	92	Two year	
9	1010	31	12	60	Month-to-month	
10	1011	27	8	68	Month-to-month	
11	1012	56	60	105	Two year	
12	1013	44	28	82	One year	
13	1014	36	20	75	One year	
14	1015	30	15	58	Month-to-month	
15	1016	47	40	90	Two year	
16	1017	22	4	45	Month-to-month	
17	1018	39	32	80	One year	
18	1019	53	54	100	Two year	

19 1020 33 16 70 Month-to-month

	InternetService	TechSupport	PaymentMethod	Churn
0	Fiber optic	No	Electronic check	Yes
1	DSL	Yes	Bank transfer	No
2	Fiber optic	Yes	Credit card	No
3	DSL	No	Electronic check	Yes
4	Fiber optic	Yes	Credit card	No
5	DSL	Yes	Bank transfer	No
6	No	No	Mailed check	Yes
7	Fiber optic	Yes	Electronic check	No
8	Fiber optic	Yes	Credit card	No
9	DSL	No	Mailed check	Yes
10	Fiber optic	No	Electronic check	Yes
11	No	Yes	Bank transfer	No
12	DSL	Yes	Credit card	No
13	Fiber optic	No	Electronic check	Yes
14	DSL	No	Mailed check	Yes
15	Fiber optic	Yes	Credit card	No
16	No	No	Mailed check	Yes
17	DSL	Yes	Bank transfer	No
18	Fiber optic	Yes	Credit card	No
19	DSL	No	Electronic check	Yes

1.2 1. Line Plot

A line plot is used to visualize trends and changes in data.

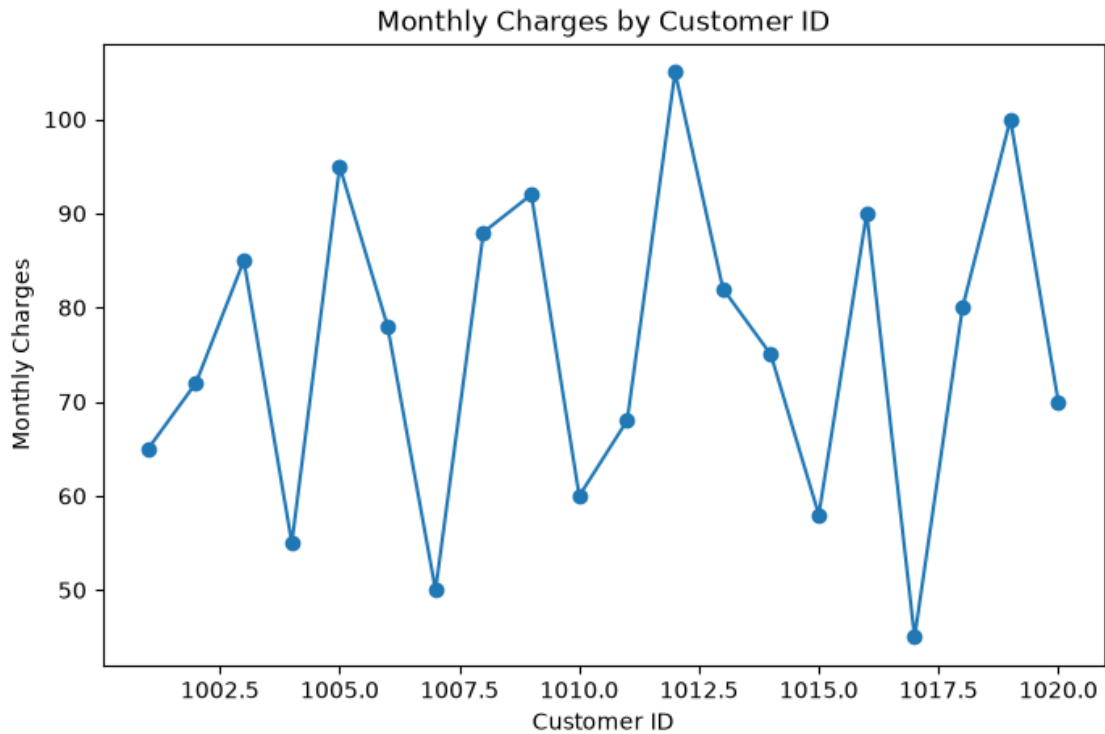
In this task, the Line Plot is used to show how Monthly Charges vary across Customer IDs.

```
[8]: plt.figure(figsize=(8, 5))

plt.plot(df["CustomerID"], df["MonthlyCharges"], marker="o")

plt.title("Monthly Charges by Customer ID")
plt.xlabel("Customer ID")
plt.ylabel("Monthly Charges")

plt.show()
```



1.3 2. Bar Plot

A bar plot is used to compare values across different categories.

In this task, the Bar Plot is used to compare the number of customers in each Contract category.

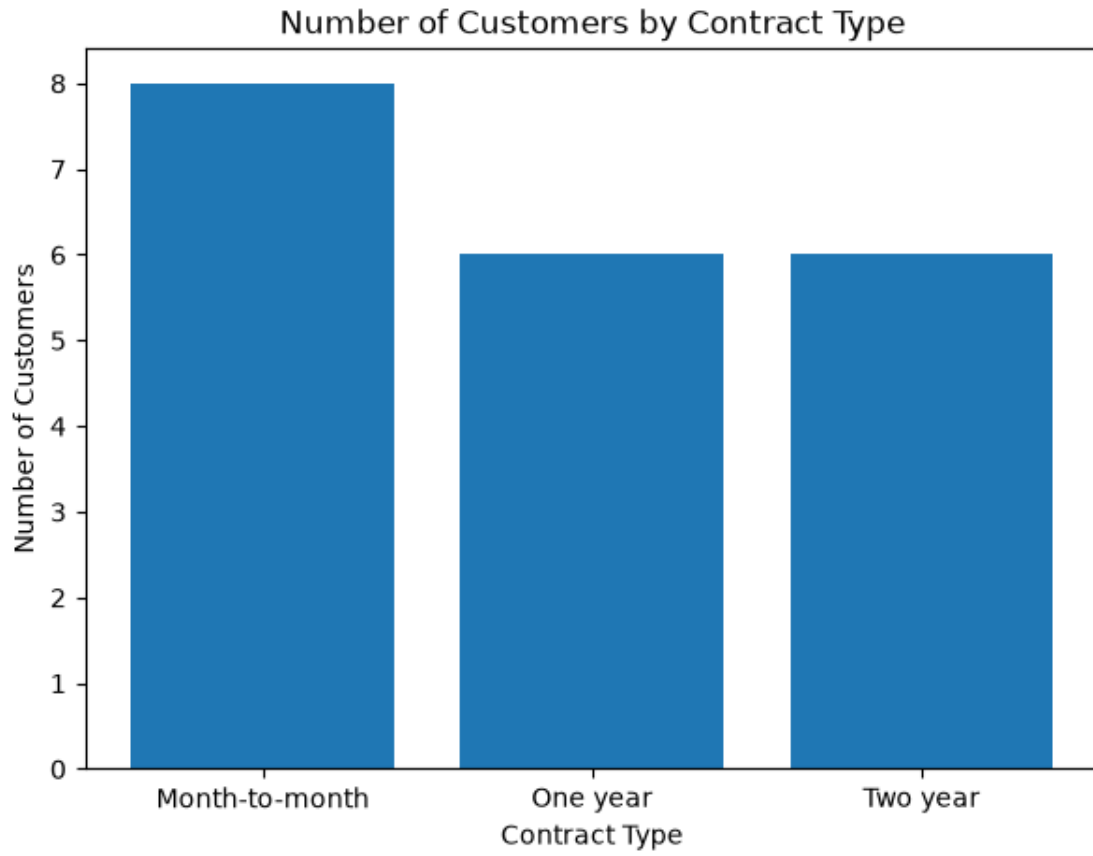
```
[9]: contract_counts = df["Contract"].value_counts()

plt.figure(figsize=(7, 5))

plt.bar(contract_counts.index, contract_counts.values)

plt.title("Number of Customers by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")

plt.show()
```



1.4 3. Histogram

A histogram is used to understand the distribution of numerical data.

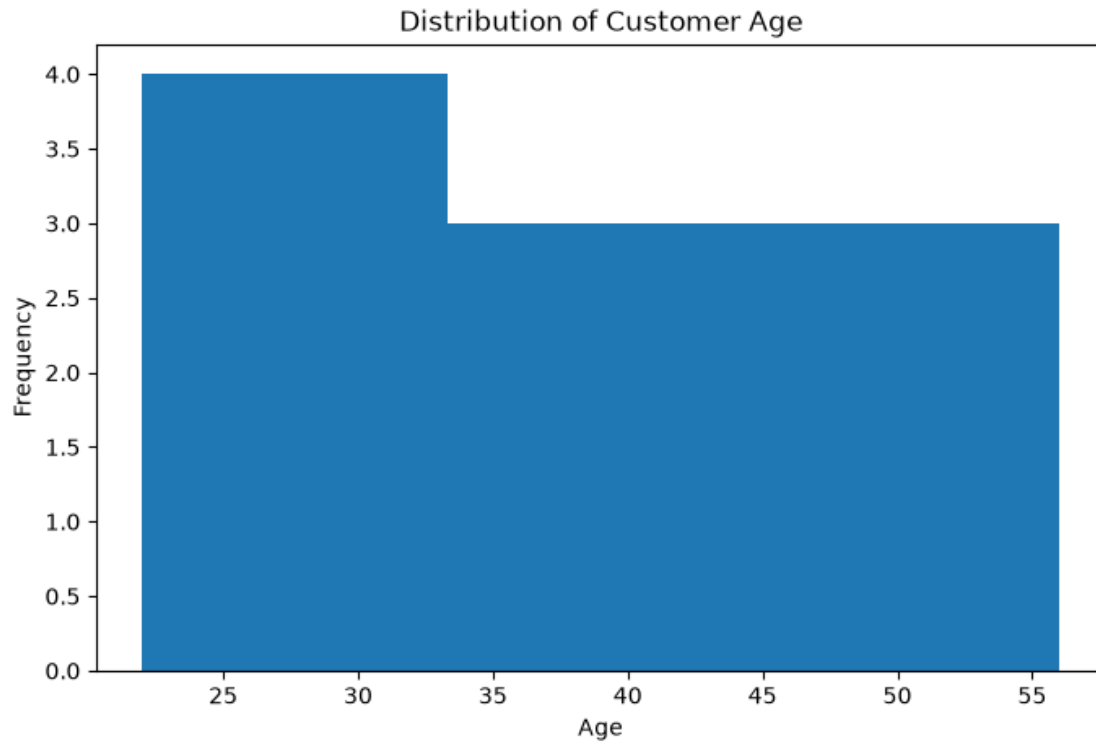
In this task, the Histogram is used to visualize the distribution of customer ages.

```
[10]: plt.figure(figsize=(8, 5))

plt.hist(df["Age"], bins=6)

plt.title("Distribution of Customer Age")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```



1.5 4. Scatter Plot

A scatter plot is used to visualize the relationship between two numerical variables.

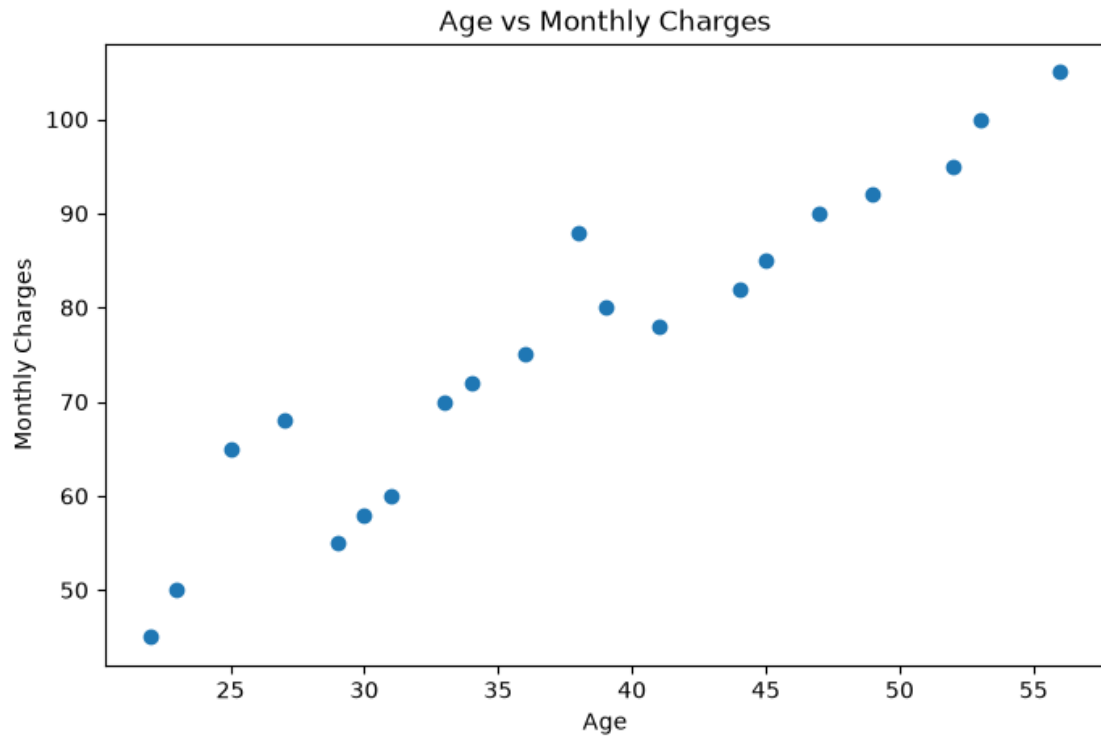
In this task, the Scatter Plot is used to understand the relationship between Age and Monthly Charges.

```
[11]: plt.figure(figsize=(8, 5))

plt.scatter(df["Age"], df["MonthlyCharges"])

plt.title("Age vs Monthly Charges")
plt.xlabel("Age")
plt.ylabel("Monthly Charges")

plt.show()
```



1.6 Conclusion

In this task, different visualization techniques were performed using Matplotlib.

A Line Plot was used to understand trends, a Bar Plot was used for categorical comparison, a Histogram was used to understand data distribution, and a Scatter Plot was used to visualize the relationship between two numerical variables.

These visualizations help in understanding the dataset more clearly and support further data analysis.